



Fuzzy Logic & Neural Networks (CS-514)

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Development of Membership Functions

- Membership functions characterize the fuzziness in a fuzzy set.
- The procedures to assign membership values to the membership functions are as follows:

1. Intuition
2. Inference
3. Rank ordering
4. Neural networks
5. Genetic algorithms
6. Inductive reasoning

Development of Membership Functions

Intuition: It is simply derived from the capacity of humans through their own intelligence and understanding.

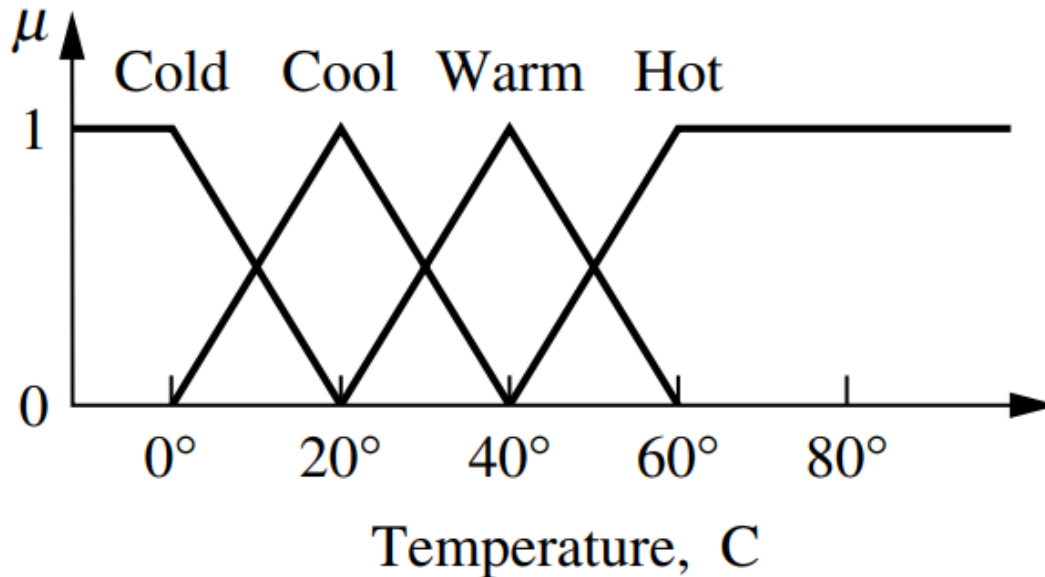


Fig.: Membership functions for the fuzzy variable "temperature."

Development of Membership Functions

Inference: In the inference method we use knowledge to perform deductive reasoning. That is, we wish to deduce or infer a conclusion, given a body of facts and knowledge.

Example: In the identification of a triangle, let A , B , and C be the inner angles of a triangle, in the order $A \geq B \geq C \geq 0$, and let U be the universe of triangles as:

$$U = \{(A, B, C) \mid A \geq B \geq C \geq 0; A + B + C = 180^\circ\}$$

Development of Membership Functions

Inference:

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For a fuzzy right angle triangle, the membership value is as:

$$\mu_{\tilde{R}}(A, B, C) = 1 - \frac{1}{90^\circ} |A - 90^\circ|$$

Development of Membership Functions

Inference:

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For the case of a fuzzy equilateral triangle, the membership function is given by

$$\mu_{\tilde{E}}(A, B, C) = 1 - \frac{1}{180^\circ}(A - C)$$

Development of Membership Functions

Inference:

Example: In the identification of a triangle, let A , B , and C be the inner angles of a triangle, in the order $A \geq B \geq C \geq 0$, and let U be the universe of triangles as:

$$U = \{(A, B, C) \mid A \geq B \geq C \geq 0; A + B + C = 180^\circ\}$$

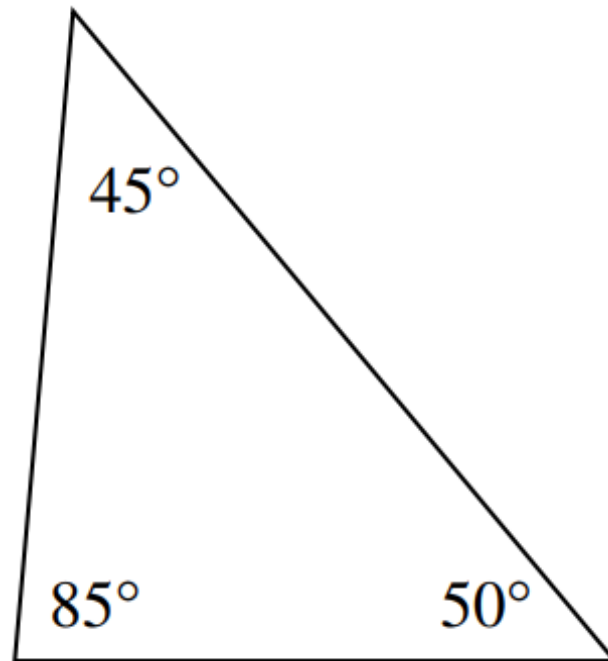
For the isosceles triangle we have the membership values as:

$$\mu_{\underline{I}}(A, B, C) = 1 - \frac{1}{60^\circ} \min(A - B, B - C)$$

Development of Membership Functions

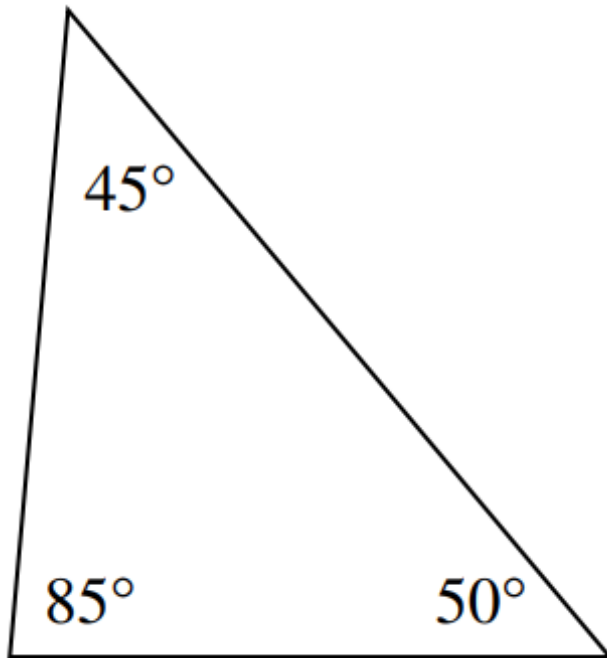
Inference:

Example: The membership values for the fuzzy triangle shown in Fig with $U = \{(A, B, C) \mid A \geq B \geq C \geq 0; A + B + C = 180^\circ\}$



Development of Membership Functions

Inference Example:



$$\mu_{\tilde{R}}(x) = 0.94$$

$$\mu_{\tilde{I}}(x) = 0.916$$

$$\mu_{\tilde{E}}(x) = 0.7$$

Development of Membership Functions

Rank Ordering: Assessing preferences by a single individual, a committee, a poll, and other opinion methods can be used to assign membership values to a fuzzy variable.

Example: Suppose 1000 people respond to a questionnaire about their pairwise preferences among five colors, $X = \{\text{red, orange, yellow, green, blue}\}$. Define a fuzzy set A on the universe of colors 'best color.'

Development of Membership Functions

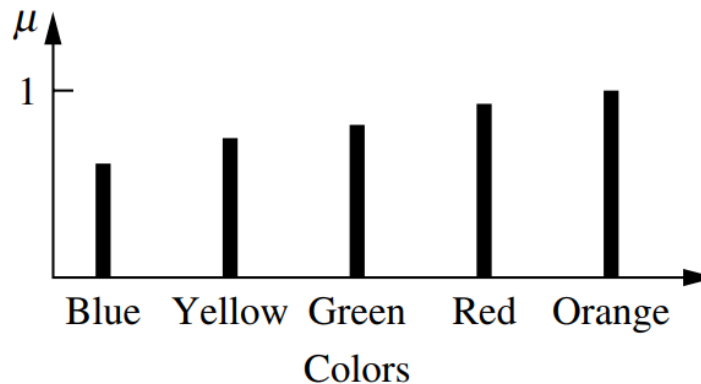
Rank Ordering Example:

$X = \{\text{red, orange, yellow, green, blue}\}$

Example in Rank Ordering

	Number who preferred					Total	Percentage	Rank order
	Red	Orange	Yellow	Green	Blue			
Red	—	517	525	545	661	2248	22.5	2
Orange	483	—	841	477	576	2377	23.8	1
Yellow	475	159	—	534	614	1782	17.8	4
Green	455	523	466	—	643	2087	20.9	3
Blue	339	424	386	357	—	1506	15	5
Total						10,000		

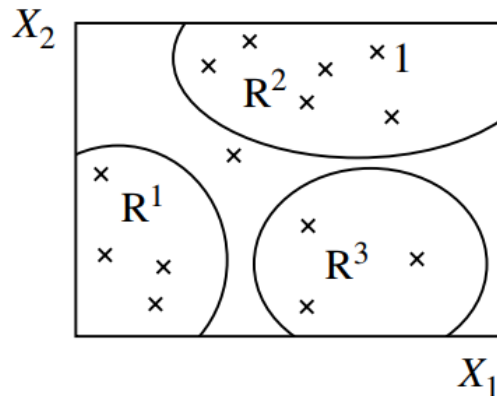
Fig.: Membership Values for “best color”



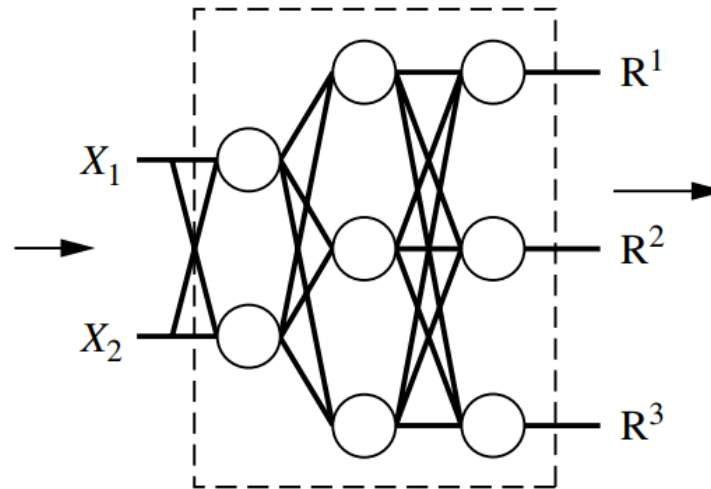
Development of Membership Functions

Neural Network: A method by which fuzzy membership functions may be created for fuzzy classes of an input data set.

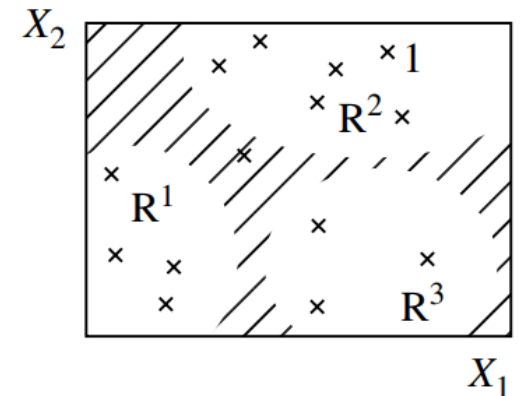
Example:



(a)



(b)



(c)

Development of Membership Functions

Genetic Algorithm: The process involves encoding membership function parameters as individuals in a population, evolving them through a genetic algorithm, and optimizing them based on a fitness function.

Example: Optimizing a Triangular Membership Function:
Steps

1. Define Membership Function
2. Select Chromosome as Membership Function Parameters.
3. Evaluate Fitness Function
4. Update GA Operators
5. Perform Iterative Optimization

Development of Membership Functions

Inductive Reasoning: It involves determining membership values or functions based on patterns or generalizations derived from observed data.

Steps for Fuzzy Membership Assignment using Inductive Reasoning:

1. Data Collection and Preprocessing
2. Initial Partitioning.
3. Assigning Membership Values.
4. Construct Membership Functions
5. Refinement using Data Patterns
6. Validation

Fuzzy c-means Clustering

- Clustering is an unsupervised machine learning technique that divides the given data into different clusters based on their distances (similarity) from each other.
- The unsupervised c-means clustering algorithm gives the values of any point lying in some particular cluster to be either as 0 or 1.
- Fuzzy c-means Clustering is a type of clustering algorithm in machine learning that allows a data point to belong to more than one cluster with different degrees of membership.

Fuzzy c-means Clustering

Example: To demonstrate the Fuzzy c-means clustering, we will consider the following data points.

Let the given data points are $\{(1, 3), (2, 5), (4, 8), (7, 9)\}$

Step 1: Initialize the data points into the desired number of clusters randomly.

Let we have two clusters

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1)	0.8	0.7	0.2	0.1
2)	0.2	0.3	0.8	0.9

Fuzzy c-means Clustering

Step 2: Find out the centroid

$$V_{ij} = \frac{\sum_{k=1}^K \mu_{ik}^m \times x_{kj}}{\sum_{k=1}^K \mu_{ik}^m}$$

Where

i is the cluster number ($i = 1, 2$)

V_{ij} is the i^{th} centroid for j^{th} index

j is the data point index ($j = 1, 2$)

k is the data point ($k = 1, 2, 3, 4; K = 4$)

μ_{ik} is the membership value of k^{th} data point for i^{th} cluster

x_{kj} is the j^{th} index of the k^{th} data point

m is the fuzziness parameter ($m = 2$)

Fuzzy c-means Clustering

Step 2: Find out the centroid.

$$V_{11} = (0.8^2 * 1 + 0.7^2 * 2 + 0.2^2 * 4 + 0.1^2 * 7) / (0.8^2 + 0.7^2 + 0.2^2 + 0.1^2) = 1.568$$

$$V_{12} = (0.8^2 * 3 + 0.7^2 * 5 + 0.2^2 * 8 + 0.1^2 * 9) / (0.8^2 + 0.7^2 + 0.2^2 + 0.1^2) = 4.051$$

$$V_{21} = (0.2^2 * 1 + 0.3^2 * 2 + 0.8^2 * 4 + 0.9^2 * 7) / (0.2^2 + 0.3^2 + 0.8^2 + 0.9^2) = 5.35$$

$$V_{22} = (0.2^2 * 3 + 0.3^2 * 5 + 0.8^2 * 8 + 0.9^2 * 9) / (0.2^2 + 0.3^2 + 0.8^2 + 0.9^2) = 8.215$$

Centroids are: (1.568, 4.051) and (5.35, 8.215)

Fuzzy c-means Clustering

Step 3: Find out the distance of each point from the centroid.

$$D_{ki} = \sqrt{\sum_{j=1}^J (x_{kj} - V_{ij})^2}$$

D_{ki} is the distance of k^{th} data point from i^{th} centroid

J is the total values in a data point (J = 2)

$$\begin{aligned} D_{11} &= ((1 - 1.568)^2 + (3 - 4.051)^2)^{0.5} \\ &= 1.2 \end{aligned}$$

$$\begin{aligned} D_{12} &= ((1 - 5.35)^2 + (3 - 8.215)^2)^{0.5} \\ &= 6.79 \end{aligned}$$

Fuzzy c-means Clustering

Step 4: Updating membership values

$$\mu_{ki} = \left[\sum_{j=1}^J \left(\frac{D_{ki}^2}{D_{kj}^2} \right)^{\frac{1}{(m-1)}} \right]^{-1}$$

For first data point the updated membership values are:

$$\mu_{11} = [\{ [(1.2)^2 / (1.2)^2] + [(1.2)^2 / (6.79)^2] \} \wedge \{ (1 / (2 - 1)) \}]^{-1} = 0.96$$

$$\mu_{12} = [\{ [(6.79)^2 / (6.79)^2] + [(6.79)^2 / (1.2)^2] \} \wedge \{ (1 / (2 - 1)) \}]^{-1} = 0.04$$

The steps 2-4 will be repeated until the values are changing significantly.

Automated Methods for Fuzzy Systems

- Automated methods for Fuzzy systems are the approaches, which are very practical and can be used to develop a model for the system using the “limited” available information. Some of these methods are:

1. Batch least squares
2. Recursive least squares
3. Gradient method
4. Clustering method
5. Learning from example (LFE)
6. Modified learning from example (MLFE)

Automated Methods for Fuzzy Systems

Batch Least Squares Method (BLS)

- This method constructs a fuzzy model from numerical data which can then be used to predict outputs given any input.
- When using the BLS algorithm to develop a fuzzy model it is helpful to have knowledge about the behaviour of the data set in order to form a rule-base.
- The accuracy of the fuzzy model developed using BLS primarily depends on the rules specified in the rule-base and the data set used to train the fuzzy model.

Automated Methods for Fuzzy Systems

Recursive Least Squares Method (RLS)

- The RLS algorithm is very similar to the BLS algorithm; however, the RLS algorithm is a recursive version of the BLS method that makes updating parameters much easier.
- The RLS algorithm requires that the rule-base be specified, (i.e., number of rules, input & output membership functions).
- If we do not have enough knowledge to define the rule-base then another algorithm should be used first.

Automated Methods for Fuzzy Systems

Gradient Method (GM)

- The GM provides a means for tuning the parameters of the fuzzy model i.e., the parameters of the rule-base.
- Not only, the input parameters the GM provides a method to tune the output membership functions as well.
- The algorithm requires that initial values for the rules be designated, but these rules are updated through the iterations with each time step.

Automated Methods for Fuzzy Systems

Clustering Method (CM)

- Fuzzy clustering is the partitioning of a collection of data into fuzzy subsets or clusters based on similarities between the data.
- The clustering method (CM), develops a fuzzy estimation model, to predict the output given the input.
- The algorithm forms rules (or clusters) with training data using a nearest neighbour approach for the fuzzy system.

Automated Methods for Fuzzy Systems

Learning from example (LFE)

- The LFE training procedure relies entirely on a complete specification of the membership functions by the analyst and it only constructs the rules.

Modified learning from example (MLFE)

- Unlike the LFE algorithm which relies entirely on user-specified membership functions, the MLFE calculates both membership functions and rules.